

Aakash Rayamajhi

Software Engineer | Backend & Distributed Systems

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TECHNICAL SKILLS

Languages: JavaScript, Python, SQL, C++, C

Backend & Distributed Systems: Node.js, Express.js, REST APIs, GraphQL, WebSockets, Webhooks, Apache Kafka, ZooKeeper, Background Workers

Databases, Cloud & DevOps: PostgreSQL, MongoDB, Redis, Elasticsearch | AWS, Docker, Kubernetes, Vercel

Frontend, AI & Data: React, React Native, Tailwind CSS | OpenAI APIs, LangChain, RAG, n8n, Pandas

Monitoring, Testing & Agile: Prometheus, Grafana, JMeter, k6 | Jira, Agile/Scrum

SELECTED PROJECTS

SkySearch – Personal Search Engine

Distributed Search System

- **Built a 3-stage search pipeline spanning crawling, indexing, and ranking**, by separating workloads into microservices for independent horizontal scaling.
- **Eliminated redundant web crawling operations** by engineering a 2-stage deduplication pipeline using URL normalization and in-memory Redis tracking prior to Kafka queuing.
- **Decoupled asynchronous processing across core services** by building Apache Kafka message queues between crawler, indexing, and ranking components to enforce fault isolation.
- **Implemented sub-second document relevance scoring** by building an inverted index hybrid search engine integrating BM25 lexical scoring and PageRank link analysis.
- **Supported scalable document retrieval** by deploying Elasticsearch for fast search indexing alongside background workers for async payload processing.

Technologies: JavaScript, Node.js, Express.js, React, Apache Kafka, Elasticsearch, Redis, PostgreSQL, Docker, BM25, PageRank, Inverted Index

e-Mistiri – Vehicle Servicing Platform

Microservices / Backend Systems

- **Built an independently deployable microservices architecture** by partitioning vehicle servicing operations into dedicated backend services with isolated data stores.
- **Accelerated geospatial query performance** by combining Geohashing with PostgreSQL spatial indexes to pre-filter candidate garages prior to detailed evaluation.
- **Separated search workloads from relational transactions** by integrating Elasticsearch for location search alongside PostgreSQL for structured relational data.
- **Eliminated race conditions on concurrent bookings** by engineering distributed mutual-exclusion locks using Redis and Apache ZooKeeper with automated TTL expiry.
- **Enhanced backend failure resiliency** by establishing asynchronous message queues, circuit breaker patterns, and reverse-proxy routing across backend components.

Technologies: JavaScript, Node.js, Express.js, React, MongoDB, PostgreSQL, Elasticsearch, Redis, ZooKeeper, Docker, Microservices, Geohashing, Message Queues

LEADERSHIP & ACHIEVEMENTS

- **Solved 300+ LeetCode, 500+ total problems** across LeetCode, HackerRank, and Codeforces, demonstrating sustained algorithmic problem-solving ability.
- **National Hackathon Runner-up** – Engineered AI features including speech transcription, automated translation, RAG-based QA, and conversational agents.
- **Nepathon Advisor** – Supported technical execution and infrastructure planning for a national hackathon, assisting participant engineering teams.
- **Competitive Free Fire In-Game Leader (2019–2022)** – Represented Nepal in international esports tournaments, leading multi-member teams through qualification rounds.

EDUCATION

NePathya College

2022–2026

B.Sc. Computer Science and Information Technology (BSc CSIT)

Tribhuvan University

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Artificial Intelligence, Computer Architecture